Hypothesis Testing with E-Values

Chapter 1: Introduction

Based on Ramdas & Wang

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Chapter Overview

Notation You'll Need for This Chapter I

Before diving in, here is a plain-English glossary of every non-elementary term and symbol used later in this deck. Nothing here requires measure theory — treat each analogy as the working definition you need.

- **Null hypothesis,** $\mathcal{P}$: the "boring" explanation of the data we test against – a *set* of candidate probability distributions (e.g., "the coin is fair," i.e., $\{\text{Bernoulli}(0.5)\}$).
- **Alternative hypothesis,** $\mathcal{Q}$: the "interesting" explanation we'd like evidence for (e.g., "the coin is biased towards heads").
- $\mathbb{P}, \mathbb{Q}$: a single probability distribution; $\mathbb{P} \in \mathcal{P}$ means $\mathbb{P}$ is one specific member of the null hypothesis set.
- **Simple vs. composite hypothesis**: simple means the set has exactly one distribution (e.g., $\{\mathbb{P}\}$); composite means it has more than one (e.g., "$\theta \neq 0.5$" covers infinitely many biased coins).
- **Random variable** X: a quantity whose value depends on the random outcome (here: our data).
- $\mathbb{E}^{\mathbb{P}}[X]$: the expectation (long-run average) of X, computed *assuming* $\mathbb{P}$ is the true distribution.

Notation You'll Need for This Chapter II

- **E-variable / e-value**: a nonnegative random variable E with $\mathbb{E}^{\mathbb{P}}[E] \leq 1$ under every null distribution. Analogy: a bet that is never favorable to you (on average) if the null is true, but pays off big if the null is false. "E-value" is the number you actually observe.
- **P-variable / p-value**: you already know this one — a number in $[0, 1]$ that is uniform-or-larger under the null. Small p-value $=$ evidence against the null. (*Opposite* convention to e-values: for e-values, *large* is evidence against the null.)
- **Test** ϕ: a (possibly randomized) decision rule taking values in $[0, 1]$, interpreted as "the probability I reject H_0." A **binary** test only ever outputs 0 ("don't reject") or 1 ("reject").
- **Type-I error / level** α: the chance of a false alarm (rejecting a true null). A "level-α test" controls this chance to be at most α (e.g., $\alpha = 0.05$).
- **Likelihood ratio** $\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P}$: for a data point x, how much more likely x is under $\mathbb{Q}$ than under $\mathbb{P}$. If data are continuous/discrete with densities (or probability mass functions) $q(x), p(x)$, this is just $q(x)/p(x)$.

Notation You'll Need for This Chapter III

- **Exact e/p-variable**: one where the defining inequality is always an equality ($\mathbb{E}^{\mathbb{P}}[E] = 1$, or $\mathbb{P}(P \leq \alpha) = \alpha$) — the "tightest possible" case.
- **Stochastically larger**: random variable X is stochastically larger than a reference distribution if X is, loosely, "at least as likely to be big." (Formally: its cdf is never above the reference cdf.)
- **Supermartingale idea (informal)**: a wealth process that, on average, never grows from one step to the next given the past — like a casino game that is fair or tilted against you. We will only need the intuition, not the formal machinery.
- **Filtration / adapted / stopping time (informal)**: think of a filtration as "the information you've seen up to now, growing over time"; a process is adapted if, at each time, it only uses information seen so far; a stopping time is a rule for "when to stop" that only looks at data observed up to that moment (no peeking into the future).

Notation You'll Need for This Chapter IV

- **E-process**: a sequence of e-values $E_1, E_2, \dots$ (built up as data arrive) that remains a valid e-value *even if you stop early*, at any data-dependent stopping time. This is what lets e-values handle "peeking" safely.
- **Exchangeable**: a collection of random variables whose joint behavior does not change if you relabel/permute them – e.g., shuffling a deck of otherwise-identical-looking cards.
- **Calibrator**: a fixed recipe that converts one kind of statistic into another while keeping validity, e.g., turning an e-value E into a p-value via $P = 1/E$.
- **Log-optimal e-value / e-power**: among all valid e-values, the one that maximizes $\mathbb{E}^{\mathbb{Q}}[\log E]$ (expected log growth rate under the alternative $\mathbb{Q}$) – explained with concrete intuition in Section 1.2.
- **Markov's inequality (used implicitly)**: for any nonnegative random variable E with $\mathbb{E}[E] \leq 1$, we automatically get $\mathbb{P}(E \geq 1/\alpha) \leq \alpha$ – this is exactly why "reject when $E > 1/\alpha$" is a valid level-α test.

1.1 The Three Roles of E-Values I

An e-value (nonnegative statistic with expectation at most 1 under the null; reminder from the notation frame) plays three distinct roles in the book. Understanding these upfront will help you see why so many later chapters revisit the same simple object.

Defining property (restated)

An e-value is a nonnegative test statistic whose expected value is at most one under the null hypothesis.

- "e" can stand for *expectation* (its defining constraint) or *evidence* (how we use it).
- Larger e-value $\Rightarrow$ more evidence *against* the null – the opposite direction from p-values.

1.1 The Three Roles of E-Values II

1. **Methodological.** E-values directly yield familiar tools: the reciprocal $1/E$ of an e-value is a valid p-value, and rejecting the null when E exceeds $1/\alpha$ is a valid level-α test (both via Markov's inequality). E-values also unlock things p-values struggle with: universal inference for irregular problems, log-optimal ("uniformly most e-powerful") e-values, sequential anytime-valid testing, and simple false-discovery-rate control.
2. **Technical.** E-values show up as intermediate tools even in problems that don't look like they involve e-values at all – e.g., every procedure that controls the false discovery rate turns out to be an instance of an e-value-based procedure in disguise. They are also intimately tied to nonnegative (super)martingales (the "fortune process never grows on average" idea from the notation frame).
3. **Fundamental.** At the deepest level: a powered test exists for a testing problem if and only if a powered e-variable exists. E-values are not just a convenience – they are equivalent, foundational objects to hypothesis tests themselves.

The book positions itself as building *a theory of evidence* (via e-values), complementing the classical *theory of decision-making* (Neyman, Pearson, Wald, Savage).

1.2.1 P-values and E-values Are Statistical Cousins I

The book does *not* argue e-values should replace p-values. Instead, it argues both deserve equal footing as complementary tools. Why call them "cousins"? Because they can be converted into each other while preserving key guarantees.

1.2.1 P-values and E-values Are Statistical Cousins II

E-to-p and p-to-e calibration

The intuition: an e-value's reciprocal behaves like a p-value, and an indicator of "was the p-value small" behaves like an e-value.

$$\text{(e-to-p)} \quad P = \frac{1}{E} \qquad\qquad \text{(p-to-e, at level } \alpha\text{)} \quad E = \frac{\mathbf{1}\{P \le \alpha\}}{\alpha}.$$

- E: an e-variable for $\mathcal{P}$; P: a p-variable for $\mathcal{P}$.
- $\mathbf{1}\{P \le \alpha\}$: the indicator function, 1 if $P \le \alpha$ is true, else 0.
- e-to-p calibration *preserves the rejection decision exactly*: for any α, $1/E \le \alpha \iff E \ge 1/\alpha$.
- The specific p-to-e calibrator above also preserves the decision at that fixed level α (it is the "all-or-nothing" e-value).

1.2.1 P-values and E-values Are Statistical Cousins III

Key differences (why you might prefer one over the other):

- **Post-hoc decisions.** With e-values, the rejection level α can be chosen *after* seeing the data, and the guarantee still holds. This is provably false for p-values in general.
- **Optional continuation.** If several scientists sequentially run follow-up experiments, their e-values can always be *merged by simple multiplication* – even if later experiments were only run because earlier ones looked promising. Sequentially-obtained p-values can be very hard to combine correctly in this setting.
- **Unknown sampling scheme.** P-values need a fully specified data-collection design to be valid; a specific family of e-values remains valid regardless of the (reasonable) stopping/design rule used – illustrated concretely in Example 1.4 (Section 1.6).
- **Multiple testing.** "Compound" e-values turn out to be central to multiple-testing procedures (Chapters 9, 11, 13) in ways that have no clean p-value analogue.

1.2.2 Power and E-power I

A natural question: *are e-values more or less powerful than p-values?* The book's answer is subtle – power is a property of a *procedure* (a threshold test), not of e-values or p-values in isolation.

- If a p-value P is derived from an e-value E via $P = 1/E$, the two thresholded tests are identical – no difference in power.
- If an e-value is derived from a p-value via $E = \mathbf{1}\{P \leq \alpha\}/\alpha$, the resulting level-$\alpha$ test again has exactly the same power as the original p-value test.
- Otherwise (general e-values vs. general p-values), no universal comparison is possible – it depends on which specific e-value or p-value you picked.

1.2.2 Power and E-power II

E-power: the criterion the book actually optimizes

Intuition first: if we care about the null hypothesis being *wrong* (alternative $\mathbb{Q}$ true), we'd like our e-value to grow as fast as possible under $\mathbb{Q}$. Taking the expected *logarithm* (rather than the expectation) turns out to be the right notion of "fast growth" when e-values will be multiplied together across repeated experiments.

$$\text{e-power of } E \text{ against } \mathbb{Q} \;:=\; \mathbb{E}^{\mathbb{Q}}[\log(E)].$$

- $\mathbb{E}^{\mathbb{Q}}[\cdot]$: expectation computed as if the alternative $\mathbb{Q}$ generated the data.
- $\log$: natural logarithm.
- An e-value that maximizes e-power is called **log-optimal**.

1.2.2 Power and E-power III

- Why the log? If scientists sequentially multiply e-values $E_1, E_2, \ldots$ (valid even under data-dependent stopping/continuation), then $\prod_{i=1}^{n} E_i = \exp(n \cdot \overline{\log E}_{[n]})$, so the *rate* at which the product diverges to ∞ is governed exactly by $\mathbb{E}[\log E_i]$.
- Log-optimality is therefore not an arbitrary choice of utility function – it is forced by the structure of repeated multiplication.

1.2.2 Power and E-power IV

One subtlety worth remembering: e-values in this book are chosen to be log-optimal, *not* to maximize power at a fixed level α.

- A power-maximizing e-value degenerates into an "all-or-nothing" e-value $\mathbf{1}_A/\alpha$ for some rejection set A – but this has e-power $= -\infty$ (it equals zero with positive probability, and $\log(0) = -\infty$).
- Log-optimal e-values, by contrast, are always strictly positive (with probability 1 under $\mathbb{Q}$), so they never risk this catastrophic penalty.

1.2.2 Power and E-power V

- **Takeaway:** if you plug a log-optimal e-value directly into a decision procedure built around classical power, you may be disappointed – not because e-values are weak, but because you're comparing objects optimized for different criteria.

Python: E-to-P and P-to-E Calibration, Checked Numerically

```python
import numpy as np
rng = np.random.default_rng(0)

alpha = 0.05
n_reps = 200_000

# Start from a valid e-value E (mean <= 1 under H0) for our coin example:
# H0: theta=0.5 (fair coin), n=20 flips, LR e-value vs theta=0.7
n, p0, q0 = 20, 0.5, 0.7
S = rng.binomial(n, p0, size=n_reps)                    # data simulated under H0
E = (q0/p0)**S * ((1-q0)/(1-p0))**(n-S)                  # e-value (eq. 1.5)

P_from_E = 1.0 / E                                       # e-to-p calibration
print("mean(E) under H0          =", E.mean())           # should be <= 1
print("P(P_from_E <= alpha)      =", (P_from_E <= alpha).mean(),
      " vs alpha =", alpha)                               # should be <= alpha
print("P(E >= 1/alpha)           =", (E >= 1/alpha).mean())  # identical event

# Now go the other way: turn a p-value into an e-value at level alpha
E_from_P = (P_from_E <= alpha).astype(float) / alpha   # p-to-e calibration
print("mean(E_from_P) under H0 =", E_from_P.mean())     # should be <= 1
```

Output (excerpt): `mean(E)` $\approx$ `0.982`, `P(P_from_E` $\leq$ `0.05)` $\approx$ `0.006` $\leq$ `0.05`, `mean(E_from_P)` $\approx$ `0.006` $\leq$ `1` — exactly the two calibration guarantees from the block above, confirmed by simulation.

1.3 Mathematical Notation and Conventions I

This section fixes the formal scaffolding used throughout the book. We only need the parts relevant to the definitions in the next section – treat the rest as background color.

- Data live in a sample space Ω (all possible outcomes), equipped with a σ-algebra $\mathcal{F}$ (informally: "all the events we're allowed to assign a probability to"; see notation frame).
- $\mathcal{M}_1 =$ the set of *all* probability distributions on $(\Omega, \mathcal{F})$. Its elements are also just called "distributions."
- A random variable X is a function from Ω to $\mathbb{R}$ (or $[-\infty, \infty]$ if we explicitly allow infinite values).
- The true (but unknown) distribution generating our data X is $\mathbb{P}^\dagger \in \mathcal{M}_1$. Data can be a vector $X = (X_1, \ldots, X_n)$, possibly (but not necessarily) i.i.d.
- $X^t := (X_1, \ldots, X_t)$: shorthand for the first t data points.

Definition 1.1: Hypothesis

Intuition: a hypothesis is simply the collection of distributions that are "consistent with" a scientific claim. If the claim pins down exactly one distribution, there's no remaining uncertainty about which $\mathbb{P}$ is true (simple); otherwise several distributions are all compatible with the claim (composite).

A *hypothesis* is a set of probability measures in $\mathcal{M}_1$. A hypothesis is *simple* if it is a singleton, such as $\{\mathbb{P}\}$ and $\{\mathbb{Q}\}$. Otherwise it is *composite*.

- We always write $\mathcal{P}$ for the null hypothesis and $\mathcal{Q}$ for the alternative hypothesis; H_0 / H_1 denote the corresponding statements "$\mathbb{P}^\dagger \in \mathcal{P}$" / "$\mathbb{P}^\dagger \in \mathcal{Q}$."
- $\mathcal{P}, \mathcal{Q}$ are always assumed non-intersecting (they can't both be true).
- $\mathrm{Conv}(\mathcal{P})$: the convex hull of $\mathcal{P}$ (all probability-weighted mixtures of distributions in $\mathcal{P}$).

1.3 Mathematical Notation and Conventions III

- $\mathbb{Q}$ is *absolutely continuous* with respect to $\mathbb{P}$, written $\mathbb{Q} \ll \mathbb{P}$, if $\mathbb{P}(A) = 0 \Rightarrow \mathbb{Q}(A) = 0$ for every measurable set A. Intuition: anything $\mathbb{P}$ deems impossible is also impossible under $\mathbb{Q}$, so it makes sense to compare their densities pointwise.
- When $\mathbb{Q} \ll \mathbb{P}$, we write $\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P}$ for the likelihood ratio (reminder from the notation frame): if q, p are densities/pmfs of $\mathbb{Q}, \mathbb{P}$, then $(\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P})(x) = q(x)/p(x)$.
- X is *stochastically larger* than a distribution with cdf F if $\mathbb{P}(X \leq x) \leq F(x)$ for all $x \in \mathbb{R}$ (reminder: "at least as likely to be big").
- $\mathbb{N} = \{1, 2, \dots\}$; $\mathbb{R} = (-\infty, \infty)$; $[K] := \{1, \dots, K\}$; Δ_K is the probability simplex in $\mathbb{R}^K$: $\Delta_K = \{(v_1, \dots, v_K) \in [0,1]^K : \sum_{k=1}^K v_k = 1\}$.
- The Euler number $\exp(1)$ is denoted e (upright) to avoid clashing with the variable E used for e-values.
- "Increasing"/"decreasing" are always meant in the *non-strict* sense; $x \wedge y = \min(x, y)$, $x \vee y = \max(x, y)$.

Here is the formal heart of the chapter. Before the wall of symbols: an e-variable is a "fair-or-worse bet against the null," a p-variable is the classical p-value object, and a test is a (possibly randomized) accept/reject rule. Let $\mathcal{P} \subseteq \mathcal{M}_1$ be the null hypothesis.

Definition 1.2: E-variables, P-variables, and Tests

(i) An *e-variable* E for $\mathcal{P}$ is a $[0, \infty]$-valued random variable satisfying $\mathbb{E}^{\mathbb{P}}[E] \leq 1$ for all $\mathbb{P} \in \mathcal{P}$. It is *exact* if $\mathbb{E}^{\mathbb{P}}[E] = 1$ for all $\mathbb{P} \in \mathcal{P}$.

(ii) A *p-variable* P for $\mathcal{P}$ is a $[0, \infty)$-valued random variable satisfying $\mathbb{P}(P \leq \alpha) \leq \alpha$ for all $\alpha \in (0, 1)$ and all $\mathbb{P} \in \mathcal{P}$. It is *exact* if $\mathbb{P}(P \leq \alpha) = \alpha$ for all $\alpha, \mathbb{P}$.

(iii) A *test* ϕ is a $[0, 1]$-valued random variable. It is *binary* if its range is $\{0, 1\}$. The *type-I error (rate)* of ϕ for $\mathbb{P}$ is $\mathbb{E}^{\mathbb{P}}[\phi]$. A test has *level* $\alpha \in [0, 1]$ for $\mathcal{P}$ if its type-I error is at most α for every $\mathbb{P} \in \mathcal{P}$.

- $\mathfrak{E} = \mathfrak{E}(\mathcal{P})$: the set of *all* e-variables for $\mathcal{P}$; $\mathfrak{U} = \mathfrak{U}(\mathcal{P})$: the set of all p-variables for $\mathcal{P}$.

1.4 Definitions of E-variables, P-variables, and Tests II

Remarks worth internalizing:

1. A test allows external randomization: draw $U \sim \mathrm{Unif}(0,1)$ independent of the data, reject if $U \leq \phi$. Then $\mathbb{E}^{\mathbb{P}}[\phi]$ is exactly the unconditional rejection probability.
2. Equivalently, P is a p-variable iff P is stochastically larger than $\mathrm{Unif}(0,1)$ under every $\mathbb{P} \in \mathcal{P}$.
3. A large realized e-value is evidence *against* H_0 (because its mean is capped at 1 under the null, so a big value is "surprising"); a small realized p-value is likewise evidence against H_0.
4. We say "an e-variable for H_0" to mean "for $\mathcal{P}$."
5. Most tests we use are binary, with $\phi_0 \equiv 0$ (never reject) as the trivial level-0 test.
6. An e-variable is allowed to equal ∞ (licence to reject outright), but only on *null-sets* (events impossible under every $\mathbb{P} \in \mathcal{P}$) – otherwise $\mathbb{E}^{\mathbb{P}}[E]$ couldn't stay ≤ 1. The constant $E \equiv 1$ is always a valid (if useless) e-variable for any $\mathcal{P}$.

Fact 1.3: Useful closure properties

Intuition: these are the "legal moves" you're allowed to make with e- and p-variables without breaking validity – e.g., you can always average several e-values together, or multiply *independent* ones, and the result is still a legitimate e-variable. Let $F_{X|\mathbb{P}}$ be the cdf of X under $\mathbb{P}$.

1. (i) For any $X, \mathbb{P}$: $F_{X|\mathbb{P}}(X)$ is a p-variable for $\mathbb{P}$.
2. (ii) $\sup_{\mathbb{P}\in\mathcal{P}} F_{X|\mathbb{P}}(X)$ is a p-variable for $\mathcal{P}$.
3. (iii) If $E_1, \ldots, E_K$ are e-variables for $\mathcal{P}$, their **arithmetic average** is an e-variable for $\mathcal{P}$.
4. (iv) If $E_1, \ldots, E_K$ are e-variables for $\mathcal{P}$ and **independent** under each $\mathbb{P} \in \mathcal{P}$, their **product** is an e-variable for $\mathcal{P}$.
5. (v) If P is a p-variable and $P' \geq P$, then P' is a p-variable.
6. (vi) If E is an e-variable and $0 \leq E' \leq E$, then E' is an e-variable.

1.4 Definitions of E-variables, P-variables, and Tests IV

E-process (informal definition)

Intuition: an e-process is a whole time series of e-values that stays trustworthy *even if you decide when to stop looking based on the data itself* – crucial for sequential science.
A finite or infinite sequence $E_1, E_2, \ldots$ of e-variables (adapted to an underlying filtration; reminder: "only uses information seen so far") is called an *e-process* if, for every stopping time τ (a rule for stopping that doesn't peek into the future), E_τ is again a valid e-value.

- This innocuous-looking definition hides a deep connection to nonnegative (super)martingales, explored fully in Chapter 7.
- It is exactly the property that makes "optional continuation" (Section 1.2.1) and "unknown sampling schemes" (Example 1.4) work.

Python: Checking Fact 1.3 (Averaging and Multiplying E-values)

```python
import numpy as np
rng = np.random.default_rng(1)
n, p0, n_reps = 20, 0.5, 100_000
S = rng.binomial(n, p0, size=n_reps)      # data simulated under H0

# Two e-variables for the SAME null, vs two different alternatives
E1 = (0.7/p0)**S * (0.3/(1-p0))**(n-S)    # LR e-value vs theta=0.7
E2 = (0.6/p0)**S * (0.4/(1-p0))**(n-S)    # LR e-value vs theta=0.6
E_avg = (E1 + E2) / 2                     # Fact 1.3(iii): average
print("mean(E_avg) =", E_avg.mean())      # <= 1 ?

# Product of e-variables from INDEPENDENT rounds (Fact 1.3(iv)):
Sa = rng.binomial(10, p0, size=n_reps)    # first 10 flips
Sb = rng.binomial(10, p0, size=n_reps)    # independent next 10 flips
Ea = (0.7/p0)**Sa * (0.3/(1-p0))**(10-Sa)
Eb = (0.7/p0)**Sb * (0.3/(1-p0))**(10-Sb)
print("mean(Ea*Eb) =", (Ea * Eb).mean())  # <= 1 ?
```

Output: both means come out close to 1 (typically 0.97–1.00) and never exceed it, confirming Fact 1.3(iii)–(iv).

1.5 Betting Interpretation of E-values

E-variables have a clean interpretation as *fair or unfavorable bets* against the null. This framing is the intuitive engine behind most of the book's constructions.

The Forecaster/Skeptic game

Intuition: Skeptic is trying to make money betting against Forecaster's claim that $\mathbb{P} \in \mathcal{P}$ describes the world. An e-variable is precisely a bet that Forecaster (a rational, expected-value-maximizing party) would be willing to accept.

- Skeptic pays \$1 up front and proposes a payout function $E : \Omega \to \mathbb{R}_+$: if outcome ω occurs, Forecaster pays Skeptic $E(\omega)$ dollars.
- Forecaster accepts the contract only when it is favorable to them – i.e., when $\mathbb{E}^{\mathbb{P}}[E] \leq 1$ (Skeptic doesn't expect to profit if $\mathbb{P}$ is really true). This is *exactly* the e-variable condition.
- If Skeptic secretly believes $\mathbb{Q}$ is the truth, they will only offer this particular E if $\mathbb{E}^{\mathbb{Q}}[E] > 1$ – i.e., they expect to come out ahead.

A large realized e-value $E(\omega)$ = Skeptic made a lot of money = strong evidence that Forecaster's null was wrong.

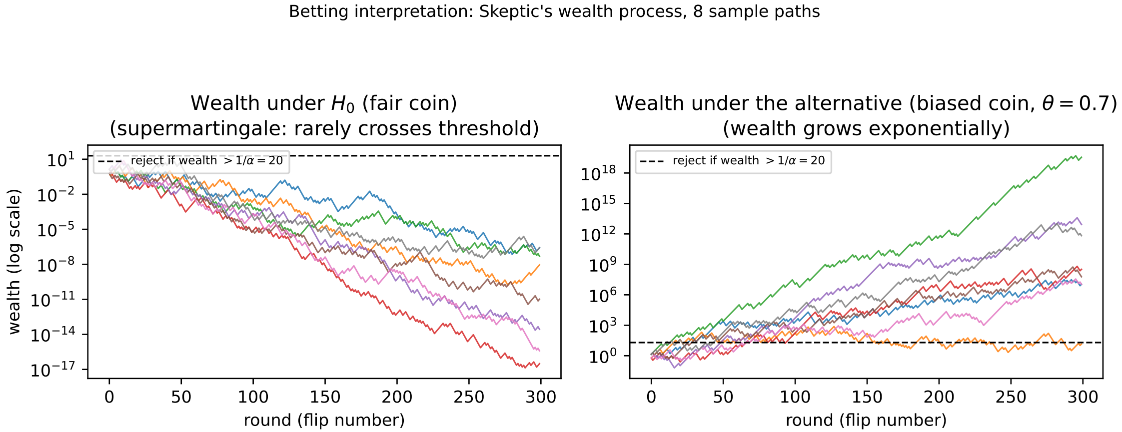
Contrast: The (Flawed) Betting Interpretation of P-values

P-values have a betting interpretation too – but a flawed one, which helps explain why e-values are the more natural currency for combining evidence across experiments.

- For every $r \in (0, 1]$, Forecaster offers a contract priced at \$1 that pays $1/r$ dollars if statistic $P < r$, and nothing otherwise.
- Forecaster only offers this if P is a p-variable for $\mathcal{P}$ (equivalently, $\mathbf{1}\{P \le r\}/r$ is an e-variable for any fixed r).
- But here, the bets are indexed by r, and the *realized* p-value equals the smallest r at which Skeptic *would have* profited, **had they picked that particular bet in advance**.
- This is "double dipping": p-values implicitly report the best bet in hindsight, whereas an e-value is a single, honestly pre-committed bet. That is why e-values and $1/(\text{p-value})$ should never be compared on the same numeric scale.

This distinction is exactly what powers the "optional continuation" and "post-hoc decision" advantages of Section 1.2.1.

Figure: The Betting Wealth Process



Eight simulated wealth paths from repeatedly betting $E = (0.7/0.5)^{\text{heads}}(0.3/0.5)^{\text{tails}}$ per flip. **Left:** under the null ($\theta = 0.5$), wealth fluctuates but rarely exceeds the rejection threshold $1/\alpha$ (supermartingale intuition: on average, wealth does not grow). **Right:** under the true alternative ($\theta = 0.7$), wealth grows exponentially – Skeptic profits.

Python: Simulating the Wealth Process and Ville's Inequality

```python
import numpy as np
rng = np.random.default_rng(2024)

alpha = 0.05
threshold = 1/alpha                # reject if wealth ever exceeds this
n_rounds = 300
n_reps = 5000
p0, q0 = 0.5, 0.7                 # H0: fair coin; bet built against theta=0.7

max_wealth = np.zeros(n_reps)
for r in range(n_reps):
    flips = rng.binomial(1, p0, size=n_rounds)          # data under H0
    per_flip_e = np.where(flips == 1, q0/p0, (1-q0)/(1-p0))
    wealth = np.cumprod(per_flip_e)                     # running product = wealth
    max_wealth[r] = wealth.max()

frac_exceeds = np.mean(max_wealth > threshold)
print(f"P(max wealth over {n_rounds} rounds > 1/alpha={threshold:.0f}) = {frac_exceeds:.4f}")
print(f"Ville's inequality guarantees this is <= alpha = {alpha}")
```

Output: `P(max wealth > 20) = 0.0490`, comfortably respecting the ≤ 0.05 guarantee – even though we checked the *running maximum* over all 300 rounds, not just the final value. This is the practical payoff of the e-process property.

1.6 A First Example: Likelihood Ratio

The most natural e-variable of all: for testing a simple null $\mathbb{P}$ against a simple alternative $\mathbb{Q} \ll \mathbb{P}$ (reminder: absolutely continuous, i.e., comparable densities), the likelihood ratio itself is an e-variable.

Likelihood-ratio e-variable

Intuition: how much more plausible is the data under $\mathbb{Q}$ than under $\mathbb{P}$? That ratio, applied to the observed data, is automatically a valid (exact) e-value for $\mathbb{P}$.

$$E = \frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}(X).$$

- X: the observed data.
- For *any* $\mathbb{Q} \ll \mathbb{P}$, this E is an e-variable for $\mathbb{P}$, and in fact *exact* ($\mathbb{E}^{\mathbb{P}}[E] = 1$).

We specialize to two settings the book uses throughout: Gaussian mean shift, and our running Bernoulli (coin flip) example.

1.6 Likelihood Ratio: the Gaussian Example

Gaussian example (Eq. 1.1–1.2)

Testing $\mathbb{P} = \mathrm{N}(0,1)$ against $\mathbb{Q} = \mathrm{N}(\mu,1)$ ($\mu > 0$ known), with i.i.d. data $X_1, \ldots, X_n$ and $S_n = \sum_i X_i$:

$$E_n = \exp\big(\mu S_n - n\mu^2/2\big) = \prod_{i=1}^{n} \exp(\mu X_i - \mu^2/2).$$

- S_n: sum of the n observations.
- Each factor $\exp(\mu X_i - \mu^2/2)$ is itself an exact e-variable; their product remains exact ($\mathbb{E}^{\mathbb{P}}[E_n] = 1$).

Writing $\delta = \mu\sqrt{n}$ (signal strength) and $Z = S_n/\sqrt{n}$ (standard normal under H_0):

$$E_n = \exp\big(\delta Z - \delta^2/2\big), \qquad P_n = \Phi(-S_n/\sqrt{n}) = \Phi(-Z)$$

- Φ: standard normal cdf; P_n is the classical Neyman–Pearson p-variable.
- E_n increasing in S_n, P_n decreasing in S_n: both agree that larger S_n favors $\mathbb{Q}$.

1.6 Likelihood Ratio: the Two-Sided Version

Two-sided version (Eq. 1.3–1.4)

Testing $\mathrm{N}(0,1)$ against $\mathrm{N}(\mu,1)$ for *unknown* $\mu \neq 0$, using a free parameter $\delta > 0$:

$$E_n = \frac{\exp(\delta Z - \delta^2/2) + \exp(-\delta Z - \delta^2/2)}{2}, \qquad P_n = 2\Phi(-|Z|).$$

- Averaging the "$+\mu$" and "$-\mu$" bets (Fact 1.3(iii)) gives a valid e-variable for the two-sided problem.
- δ is a design choice affecting power – addressed in later chapters.

1.6 Likelihood Ratio: the Bernoulli (Coin-Flip) Version

Bernoulli / coin-flip version (Eq. 1.5) – our running example

Testing $\mathbb{P} = \text{Bernoulli}(p)$ vs. $\mathbb{Q} = \text{Bernoulli}(q)$, $q > p$, sample size n, $S_n = \sum_i X_i$ (number of heads):

$$E_n = \left(\frac{q}{p}\right)^{S_n} \left(\frac{1-q}{1-p}\right)^{n-S_n} .$$

- E_n exact, increasing in S_n; Neyman–Pearson p-variable is $P_n = 1 - F_{n,p}(S_n)$ (upper binomial tail).
- This is our **running example** for the rest of the deck: testing whether a coin is fair.

Worked Example: Is Our Coin Fair? (Step by Step)

Setup. $H_0 : \theta = 0.5$ (fair coin) vs. chosen alternative $\theta = 0.7$. We flip $n = 20$ times and observe $S_{20} = 14$ heads.

Computing the e-value, one step at a time

- **Step 1** (identify the formula): use Eq. 1.5 with $p = 0.5$, $q = 0.7$, $n = 20$, $S_n = 14$.
- **Step 2** (plug in): $E_{20} = (0.7/0.5)^{14} \, (0.3/0.5)^6 = (1.4)^{14}(0.6)^6$.
- **Step 3** (compute each factor): $1.4^{14} \approx 111.12, \quad 0.6^6 = 0.046656$.
- **Step 4** (multiply): $E_{20} \approx 111.12 \times 0.046656 \approx 5.18$.
- **Step 5** (interpret): $E_{20} \approx 5.18 > 1$ is mild evidence *against* the fair-coin null – but well below $1/\alpha = 20$ for $\alpha = 0.05$, so we would *not* reject at that level.

For comparison, the exact one-sided p-value is $P(S_{20} \geq 14 \mid \theta = 0.5) \approx 0.058$ – also not significant at $\alpha = 0.05$, consistent with the e-value's verdict.

Example 1.4: Unknown Sampling Scheme I

This is the book's own example illustrating why e-values tolerate unknown data-collection designs, while p-values do not.

Setup

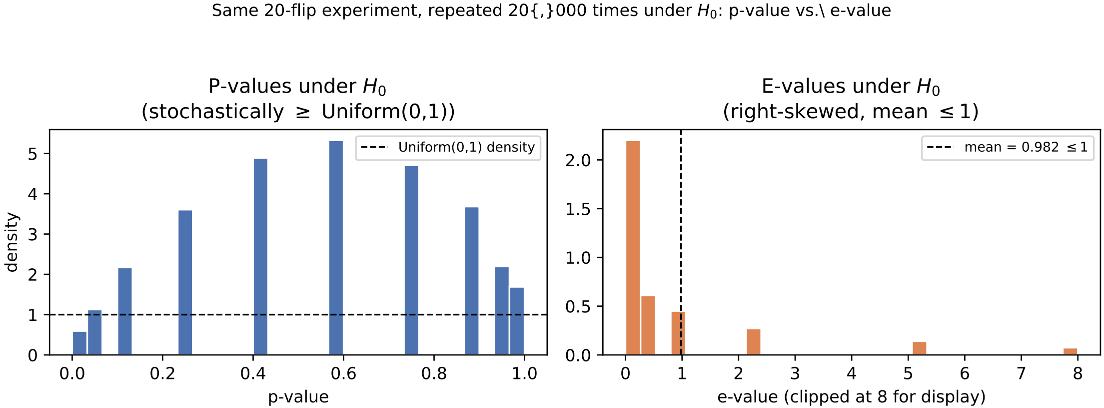
Testing $\mathbb{P} = \mathrm{Bernoulli}(1/2)$ vs. $\mathbb{Q} = \mathrm{Bernoulli}(q)$, $q > 1/2$. A scientist is handed the dataset $(1, 1, 0, 1, 1, 1, 1, 1)$ – 8 points, 7 heads – *without* being told how it was collected.

- **Design (a):** collect exactly 8 points. P-value $= \mathbb{P}(N_8 \geq 7) = 0.035$ (significant at 0.05).
- **Design (b):** collect data until 5 heads occur in a row. P-value $= \mathbb{P}($5-in-a-row observed at or before $n = 8) = 0.078$ (*not* significant at 0.05).

Example 1.4: Unknown Sampling Scheme II

- **The problem:** if only the data (not the design) is reported, the scientist cannot tell whether they're in case (a), (b), or some other design – yet the p-value's validity crucially depends on knowing which.
- **The fix:** the e-value $E_n = q^{S_n}(1-q)^{n-S_n}/2^n$ from Eq. 1.5 (with $p = 1/2$) remains a *valid* e-variable **regardless of the data-collecting design**, as long as all obtained data is used from the start.
- **Why:** $(E_n)_{n \geq 1}$ forms an e-process (Chapter 7) – it stays valid at any data-dependent stopping rule, which is exactly the "no peeking into the future" property from the notation frame.
- **Caveat:** this does not rescue *malicious* designs (cherry-picking only 1s, discarding early 0s, reporting only the "best" of several batches) – no statistical method purely based on collected data can fix that.

Figure: P-values vs. E-values Under Repeated Null Experiments



20,000 simulated repeats of our $n = 20$ coin-flip experiment under $H_0 : \theta = 0.5$. **Left:** p-values are (stochastically) uniform on $[0, 1]$ – as required by the p-variable definition. **Right:** e-values are right-skewed (most realizations are modest or small, a few are large) with sample mean $\approx 0.98 \leq 1$, exactly as the e-variable definition demands. Numerically, $\mathbb{P}(E > 20) \approx 0.006 \leq 1/20$, confirming Markov's inequality.

An Illustration: Sequential Data and Adaptive Strategies I

Suppose data $X_1, \ldots, X_n$ from parameter $\theta^\dagger$ arrive one at a time, testing $H_0 : \theta^\dagger = 0$ vs. $H_1 : \theta^\dagger \in \Theta_1$. Let $\ell(x; \theta) = \mathrm{d}\mathbb{Q}_\theta / \mathrm{d}\mathbb{P}(x)$. Any choice of θ turns $\ell(X_i; \theta)$ into a valid e-variable for H_0 – so we have freedom in how to pick θ at each step:

(a) **Fixed**: pick $\theta_1, \ldots, \theta_n \in \Theta_1$ in advance (possibly all equal).

(b) **Adaptive**: estimate θ_k from $X_1, \ldots, X_{k-1}$ only – e.g., via maximum likelihood (MLE) or maximum a posteriori (MAP).

An Illustration: Sequential Data and Adaptive Strategies II

The running product is always a valid martingale

Intuition: no matter how cleverly (or adaptively) you pick each θ_i, as long as you only use *past* data to do so, multiplying per-round e-variables together keeps you honest – your expected next-step wealth, given everything so far, is exactly your current wealth.

Define $M_n := \prod_{i=1}^{n} E_i$ with $E_i = \ell(X_i; \theta_i)$, $M_0 = 1$.

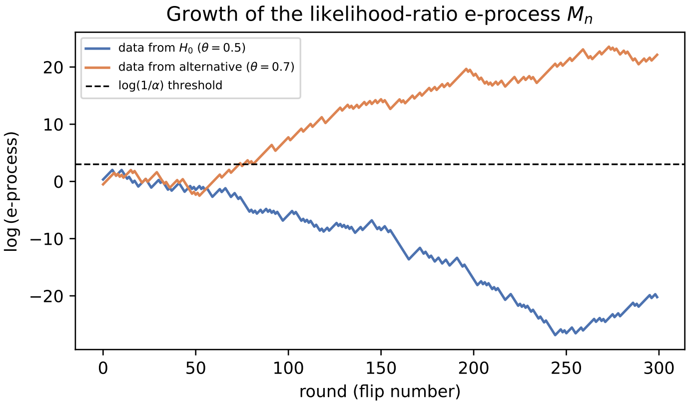
$$\mathbb{E}^{\mathbb{P}}\big[M_i \mid X_1, \ldots, X_{i-1}\big] = M_{i-1} \cdot \mathbb{E}^{\mathbb{P}}\big[\ell(X_i; \theta_i) \mid X_1, \ldots, X_{i-1}\big] = M_{i-1}.$$

- **Step 1**: factor $M_i = M_{i-1} \cdot E_i$ out of the conditional expectation (M_{i-1} is already known given the past).
- **Step 2**: θ_i is computed from $X_1, \ldots, X_{i-1}$ only, so $\mathbb{E}^{\mathbb{P}}[\ell(X_i; \theta_i) \mid X_1, \ldots, X_{i-1}] = 1$ (it's an exact e-variable given the past).
- **Conclusion**: (M_i) never grows *on average* given the past – the martingale/supermartingale property from the notation frame.

An Illustration: Sequential Data and Adaptive Strategies III

Example 1.5 (book): data from $\mathrm{N}(\theta^\dagger, 1)$ with true $\theta^\dagger = 0.3$; $H_0 : \theta^\dagger = 0$. Six strategies for choosing θ_i: (i) true value 0.3 (log-optimal); (ii) misspecified 0.1; (iii) random, $\theta_i \sim \mathrm{Unif}[0, 0.5]$ i.i.d.; (iv) MAP with prior $\mathrm{N}(0.1, 0.2^2)$; (v) MLE from $X_1, \ldots, X_{i-1}$ (started at $\theta_1 = 0.1$); (vi) discrete mixture over a uniform grid on $[0, 0.6]$. For large n, methods (iv), (v), (vi) grow *almost as fast* as the (unrealistic, oracle) method (i); misspecified/random strategies (ii), (iii) grow more slowly.

An Illustration: Sequential Data and Adaptive Strategies IV



Our own simplified illustration in the same spirit: a single e-process run under H_0 (bounded, rarely crosses the $1/\alpha$ threshold) versus under the true alternative (grows essentially linearly in log-scale, i.e. exponentially).

Python: 1000 Replications — Confirming Mean E-value ≤ 1

```python
import numpy as np
rng = np.random.default_rng(3)

n, p0, q0 = 20, 0.5, 0.7
n_reps = 1000
S = rng.binomial(n, p0, size=n_reps)           # simulate data under H0, 1000 times
E = (q0/p0)**S * ((1-q0)/(1-p0))**(n-S)         # e-value for each replication

print("Sample mean of E over 1000 replications:", E.mean())
print("Sample max of E:", E.max())
print("Fraction with E > 1/0.05 = 20:", np.mean(E > 20))

# The one specific dataset we highlighted in the worked example:
S_obs = 14
E_obs = (q0/p0)**S_obs * ((1-q0)/(1-p0))**(n-S_obs)
print("E for our observed 14-heads-out-of-20 dataset:", round(E_obs, 4))
```

Output (excerpt): `Sample mean of E` $\approx$ `0.99` (consistent with the ≤ 1 guarantee up to Monte Carlo error), `Fraction with E > 20` $\approx$ `0.005--0.01` (≤ 0.05, as Markov's inequality demands), and `E for our observed dataset = 5.1844`, matching the hand computation on the previous slide.

1.7 A Second Example: The Soft-Rank E-value

Many classical procedures (permutation tests, Monte Carlo tests) produce statistics that are *exchangeable* under the null (reminder: their joint distribution is unchanged by relabeling) and use the *rank* of the real statistic among simulated copies as a p-value. This section converts that idea into an e-value.

Setup

Intuition: we have one real test statistic L_0 and B "fake" copies $L_1, \ldots, L_B$ that look statistically identical to L_0 under the null (e.g., generated by random permutations). We want to turn the comparison "is L_0 unusually large?" into a valid e-value.

- L_0: statistic from the real data (larger $=$ more evidence against H_0).
- $L_1, \ldots, L_B$: B additional statistics, exchangeable together with L_0 under $\mathcal{P}$.
- **Example source of exchangeability**: testing independence of paired data (X_i, Y_i) via sample correlation L_0; permute the X's randomly B times to get $L_1, \ldots, L_B$.

The Soft-Rank E-value Formula

Definition (soft-rank e-value)

First transform $L_0, \ldots, L_B$ into nonnegative scores $R_0, \ldots, R_B$ that preserve their order and exchangeability (details next slide), then define

$$E = \frac{(B+1)R_0}{\sum_{b=0}^{B} R_b} \qquad \text{(with } E := 1 \text{ if } \textstyle\sum_b R_b = 0\text{)}.$$

- R_0: the transformed score of the *real* data's statistic.
- $\sum_{b=0}^{B} R_b$: total score across the real statistic and all B exchangeable copies.
- *Why this is an e-variable:* by exchangeability of $R_0, \ldots, R_B$ under $\mathcal{P}$, one can show $\mathbb{E}^{\mathbb{P}}[R_0 \mid \sum_b R_b] = \frac{\sum_b R_b}{B+1}$, which after rearranging gives $\mathbb{E}^{\mathbb{P}}[E] = 1$ (exact).

Example 1.7: A Concrete Score Transformation

Example 1.7: one concrete transformation

For a constant $r > 0$ and $L_* = \min_b L_b$:

$$R_b = \frac{\exp(rL_b) - \exp(rL_*)}{r} \geq 0.$$

- As $r \to 0$: $R_b \to L_b - L_* \geq 0$ (a simple shift).
- If the L_b are already nonnegative, the simplest valid choice is $r = 0$ with $L_* = 0$, i.e., **just use** $R_b = L_b$ **directly**.

Concrete Numbers: Soft-Rank E-value on Our Coin Data

We instantiate the abstract machinery with our own 20-flip dataset (14 heads), using the **longest run of heads** as the test statistic (a classic way to check whether a sequence looks "too streaky" to be a fair, independent coin).

Step-by-step calculation

- **Data:** `1101110101010101 0111` (14 heads, 6 tails, engineered with a run of 6 heads at the start).
- **Step 1:** compute $L_0 =$ longest head-run in the original data $= 6$.
- **Step 2:** draw $B = 4$ independent random permutations of the same 20 flips; compute longest head-runs $L_1, \ldots, L_4 = 7,\ 5,\ 4,\ 5$.
- **Step 3:** since all $L_b \geq 0$, take $R_b = L_b$ (the $r = 0$ choice). So $R_0, \ldots, R_4 = 6, 7, 5, 4, 5$, summing to 27.
- **Step 4:** soft-rank e-value $E = \dfrac{(4+1) \times 6}{27} = \dfrac{30}{27} \approx 1.111$.
- **Step 5:** ordinary permutation p-value $P = \dfrac{\#\{b : L_b \geq L_0\}}{B+1} = \dfrac{2}{5} = 0.4$ (namely $b = 0, 1$ tie/exceed $L_0 = 6$).
- **Check the book's inequality** $P \leq 1/E$: indeed $0.4 \leq 1/1.111 = 0.9$. ✓

Python: Computing the Soft-Rank E-value

```python
import numpy as np
def longest_run(seq, val=1):
    best = cur = 0
    for x in seq:
        cur = cur + 1 if x == val else 0
        best = max(best, cur)
    return best

seq = [1,1,1,1,1,1,0,1,0,1,0,1,0,1,0,1,0,1,1,1]  # 20-flip data, 14 heads
L0 = longest_run(seq)
rng = np.random.default_rng(11)
B = 4
Ls = [L0] + [longest_run(list(rng.permutation(seq))) for _ in range(B)]
R = np.array(Ls, dtype=float)            # r=0 choice: R_b = L_b (nonnegative)
E = (B + 1) * R[0] / R.sum()
P = np.mean(np.array(Ls) >= L0)          # ordinary permutation p-value
print("L_0..L_4 =", Ls, " E =", round(E,4), " P =", P, " P<=1/E:", P <= 1/E)
```

Output: `L_0..L_4 = [6, 7, 5, 4, 5]`, `E = 1.1111`, `P = 0.4`, `P<=1/E: True` $(0.4 \leq 0.9)$ – matching the hand calculation exactly.

Why "Soft-Rank"?

- The usual permutation p-value only looks at the *rank* of L_0 among $L_0, \ldots, L_B$: $P = \sum_b \mathbf{1}\{L_b \geq L_0\}/(B+1)$.
- After the exponential-style transform, $1/E$ can be seen as a "smoothed" version of that rank – hence **soft-rank p-value** for $1/E$, and **soft-rank e-value** for E (analogous to how "soft-max" smooths the maximum via exponentiation).
- The book shows algebraically that $P \leq 1/E$ always – the direct p-value is never larger than the soft-rank p-value, so at first glance there's no advantage to using E for a *single* test.
- **But:** Chapters 8–11 show e-values enjoy real advantages over p-values specifically in multiple testing and building confidence intervals – advantages invisible when testing just one hypothesis.

1.8 Historical Context: Who Invented E-values? I

Short answer: nobody, and everybody. If $\mathcal{P} = \{\mathbb{P}\}$ is simple, every optimal e-value is (or is dominated by) a likelihood ratio – and likelihood ratios are $\sim$100 years old, central to both frequentist and Bayesian statistics. What's new is recognizing e-values as a *general, composite-hypothesis-friendly* concept and the coordinated choice of the term "e-value" itself (agreed jointly by researchers in 2020).

The book credits five central historical figures, in chronological order:

- **Ville (1939):** brought (nonnegative) martingales to the center of probability theory; showed any measure-zero event has a nonnegative martingale exploding to ∞ on it. Motivated by critiquing von Mises' theory of "collectives," not statistical inference per se.
- **Wald (1945):** invented sequential analysis; nonnegative martingales (esp. the likelihood-ratio process) are central to his sequential probability ratio test, though framed around a single pre-specified stopping time.

1.8 Historical Context: Who Invented E-values? II

- **Kelly (1956):** proposed *log-optimality* – betting to maximize the expected log-growth rate of wealth in repeated favorable games, linking gambling to Shannon's information theory. (Breiman, 1961, generalized this beyond binary outcomes.)
- **Robbins (1970, with Darling 1967):** defined confidence sequences and "power-one" tests; recognized that products of e-values yield nonnegative supermartingales, and pushed for *anytime-valid* inference rather than Wald's fixed stopping rule.
- **Cover (1984):** developed log-optimal portfolio theory for financial trading with regret bounds against the best fixed portfolio in hindsight – a close cousin of log-optimal e-values.

Vovk later spearheaded the modern *game-theoretic probability* program (from 1993 onward, with Shafer and others), culminating in the term "e-value" being jointly adopted by Vovk, Shafer, Grünwald, Ramdas, and Wang around 2020.

Recent Progress: The 2018–2020 Explosion I

A remarkable concentration of independent, aligned papers appeared on arXiv within a two-year window:

- **2018 (Howard, Ramdas, McAuliffe, Sekhon):** e-processes via families of supermartingales + Ville's inequality $\Rightarrow$ nonparametric confidence sequences.
- **2018 (Kaufmann & Koolen):** mixture method for likelihood ratios in exponential families; sequential KL-divergence bounds.
- **2019 (Shafer):** expository case for betting-based scientific communication and log-optimality.
- **2019 (Shafer & Vovk, book):** game-theoretic foundations of probability and finance.
- **2019 (Grünwald, De Heide, Koolen):** simpler e-process definition (valid at any stopping time); reverse information projection; meta-analysis applications.

Recent Progress: The 2018–2020 Explosion II

- **2019 (Vovk & Wang):** averaging is essentially the *only* admissible way to combine dependent e-values; e-to-p and p-to-e calibrators. This paper triggered the community-wide agreement on the term "e-value."
- **2019 (Wasserman, Ramdas, Balakrishnan):** "universal inference" – e-values from split-likelihood-ratio statistics, valid with zero regularity conditions.
- **2020 (Vovk, Wang, Wang):** admissible p-value merging must go through e-values (via optimal transport duality).
- **2020 (Wang & Ramdas):** e-BH, an e-value analogue of Benjamini–Hochberg, controlling FDR under *arbitrary* dependence.
- **2020 (Ramdas, Ruf, Larsson, Koolen):** necessary and sufficient conditions for admissible SAVI (sequential anytime-valid inference) tools.
- **2020 (Waudby-Smith & Ramdas):** testing-by-betting for bounded-mean estimation; universal representation of nonnegative martingales.

Unfortunate Name Clashes: Other "E-values" in the Wild

The letter "e" was popular before this book – watch out for these unrelated uses of "E-value" in other literatures:

- **BLAST E-value** (bioinformatics): expected number of chance hits in a sequence database; roughly $E = -\log(1-P)$ applied to a p-value – a different object entirely.
- **Sensitivity E-value** (VanderWeele & Ding, 2017, causal inference): the minimum confounding strength (on a risk-ratio scale) needed to fully explain away an observed association.
- **Bayesian E-value** (Pereira & Stern, 1999): the *epistemic* value $ev(H \mid X)$ in the Full Bayesian Significance Test – a Bayesian evidence measure, unrelated to this book's expectation-based definition.

Reassuringly, the terms "e-variable" and "e-process" have no known clashes – use them alongside "e-value" when precision matters.

1.9 Road Map of the Book I

The book is organized into three parts, building from foundations to advanced applications.

Part I: Fundamental Concepts (Ch. 1–4)

- **Ch. 2 — Validity.** Properties of e-values under the null: Markov's inequality, calibrators, randomized tests; explicit e-variables in simple examples.
- **Ch. 3 — Efficiency.** Properties under the alternative: e-power, log-optimality, existence of powered e-values, mixture/plug-in methods.
- **Ch. 4 — Post-hoc decisions.** E-values' role when parts of the testing problem (level, decisions, or even data collection) depend on the data itself.

1.9 Road Map of the Book II

Part II: Core Ideas (Ch. 5–9)

- **Ch. 5 — Universal inference:** a simple, general method for building e-variables for composite nulls/alternatives – the first known test for many "irregular" problems (though improvable).
- **Ch. 6 — Log-optimal e-variables:** a unique "numeraire" e-variable always exists for a fixed alternative, connected to the reverse information projection.
- **Ch. 7 — SAVI:** sequential anytime-valid inference – e-processes, testing by betting, optional stopping, Ville's inequality.
- **Ch. 8 — Merging e-values:** combining via averages (arbitrary dependence) or products (independence).
- **Ch. 9 — Compound e-values:** e-Benjamini–Hochberg for false discovery rate control.

1.9 Road Map of the Book III

Part III: Advanced Topics (Ch. 10–16)

- **Ch. 10:** approximate/asymptotic e- and p-values.
- **Ch. 11:** combining dependent e-values is only admissible via (weighted) averaging.
- **Ch. 12:** combining dependent p-values (via e-values and optimal transport duality).
- **Ch. 13:** e-confidence intervals; e-Benjamini–Yekutieli for false coverage rate.
- **Ch. 14:** further e-value/p-value contrasts and dualities.
- **Ch. 15:** improved (sub-$1/\alpha$) thresholds under shape or dependence assumptions.
- **Ch. 16:** connections to risk measures; nonparametric tests for forecasts.

1.9 Road Map of the Book IV

Four cross-cutting goals

1. **Understanding the foundations:** Chapters 2, 4, 10, 14.
2. **Constructing good/powerful e-values:** Chapters 3, 6, 15.
3. **Designing inference methods:** Chapters 5, 7, 16.
4. **Designing multiple-testing methods:** Chapters 8, 9, 11, 12, 13.

In the first two categories, the object of interest is the e-value *itself*; in the remaining two, the e-value is a means to a specific statistical end.

Chapter 1 — Key Takeaways

- An **e-value** is a nonnegative statistic with expectation ≤ 1 under the null – a valid bet against H_0; large values are evidence against the null (opposite convention from p-values).
- E-values and p-values are **statistical cousins**: convertible via $P = 1/E$ and $E = \mathbf{1}\{P \leq \alpha\}/\alpha$, but e-values uniquely support post-hoc levels, optional continuation (via simple multiplication), and unknown sampling schemes.
- The preferred optimality criterion is **e-power** $\mathbb{E}^{\mathbb{Q}}[\log E]$, forced by the algebra of repeatedly multiplying e-values together.
- The **betting interpretation** (Forecaster vs. Skeptic) is the intuitive engine behind the whole framework: e-variables are exactly the wealth-generating bets Forecaster would accept.
- Two worked templates recur throughout the book: the **likelihood-ratio e-value** (our coin example: $n = 20$, 14 heads, $E \approx 5.18$) and the **soft-rank e-value** (permutation-style tests, our longest-run example: $E \approx 1.11$).
- Five names anchor the history: **Ville, Wald, Kelly, Robbins, Cover** – with Vovk, Shafer, Grünwald, Ramdas, and Wang driving the modern synthesis.
- The rest of the book builds outward from these definitions: validity (Ch. 2), efficiency (Ch. 3), post-hoc decisions (Ch. 4), and onward through sequential and multiple-testing applications.